

FlowZen: Using Hybrid-Haptic and Particle for Enhancing Immersive Experience via Continuous Illusion of Wind

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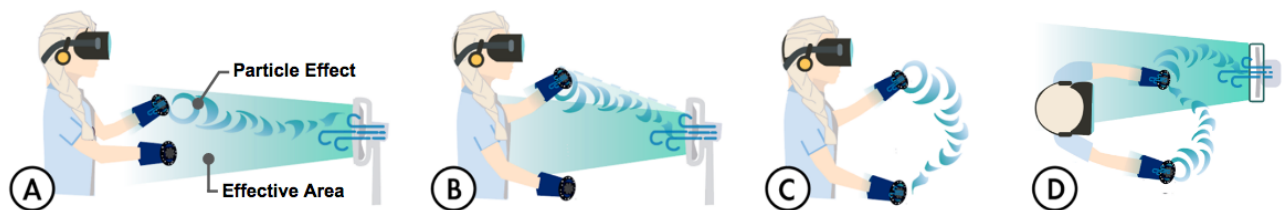


Figure 1: We introduce FlowZen, a hybrid-haptic system with particle effect for enhancing the wind sensation in the immersive environment via the tactile illusion, which includes four interaction techniques: (a) extending the effective distance of wind feedback from the stationary haptic device, (b) expanding the angle of the effective area from the stationary device, (c) bending the wind between hands, and (d) redirect the wind direction from the stationary device.

ABSTRACT

With the advancement of haptic technology, we can enable a haptic experience to simulate environmental changes seamlessly, even as the user moves around in the immersive virtual reality world. For instance, stationary non-contact tactile devices, such as electric fans or heat lights, can be utilized to replicate wind and heat sensations in the virtual environment. While these stationary devices can effectively provide a full-body experience, the impact diminishes quickly with distance. Conversely, portable devices can provide better experience, but the haptic feedback might affect only partial body areas and the direction of source is limited. Therefore, taking advantage of both portable and stationary devices to provide a complementary effect is a potential approach. We present FlowZen, a hybrid-haptic system with particle effect that enhances wind and thermal sensations through tactile illusion, integrating handheld haptic devices with contact haptic feedback and a stationary device with non-contact feedback. In our approach, visual particle effects are employed to connect the feedback from both haptic devices.

CCS CONCEPTS

• Human-centered computing → Virtual reality; Haptic devices.

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KEYWORDS

Immersive Experience, Wind Sensation, Hybrid-Haptic, Particle Effect, Tactile Illusion, Virtual Reality

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1 INTRODUCTION

Numerous studies have explored virtual reality (VR) simulations using haptic techniques to provide multi-sensory feedback for enhanced immersive experiences. To ensure haptic feedback sensations for users walking around in VR, it is necessary to use either a stationary device with a limited affected area or a portable device. Hülsmann et al.[Hülsmann et al. 2014] utilizes a fan array and a heat light array to provide wind and thermal feedback for users walking around in a three-sided CAVE. AoEs[Han et al. 2017] employ steerable multiple non-contact tactile displays to simulate thermal, wind, and humidity sensations in the virtual environment. Users can walk around in a room-scaled space with minimal hardware configurations. In contrast, Season Traveller[Ranasinghe et al. 2018] shows another approach by using a contact multi-sensory feedback device attached to VR-HMD. Although the immersion is successfully enhanced by attaching thermal sensation to the user's neck and providing wind feedback to the front of the head, it falls

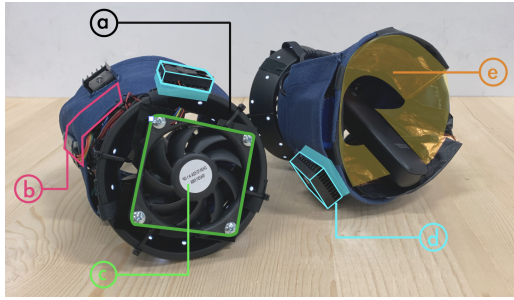


Figure 2: FlowZen hardware design: (a) VR hand controller, (b) microcontroller with Bluetooth and battery, (c) Fan, (d) Peltiers, and (e) Copper foil.

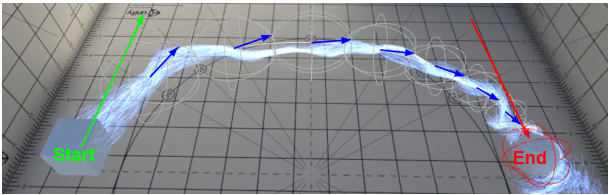


Figure 3: Our design of particle effect.

short in delivering feedback to other body parts, such as the hands. By utilizing multiple devices, advantages from one can be used to eliminate limitations from another device, leading to augmented haptic feedback. Hand-to-Hand[Pittera et al. 2017] implements two vibrating handles to simulate the intermanual illusion of movement. Haptic Around[Han et al. 2018] develops a hybrid-haptic system for enhancing interaction in an immersive environment with multiple non-contact tactile sensations. In our previous work[Hsieh et al. 2020], a handheld device is utilized to enhance wind sensations from a stationary fan, resulting in an increased perceptual area for wind. Although these works show the potential of utilizing tactile illusion, we further investigate how to coordinate multiple devices through visual effects. In this work, we design a hybrid-haptic system with particle effect to enhance the immersive experience via the continuous illusion of wind.

2 IMPLEMENTATION

Our hybrid haptic system consists of portable controllers as Figure 2 and a stationary device. Using VR hand controllers as a base, we utilize peltier and fan mechanisms to simultaneously provide contact wind and thermal feedback on the user's hands. To show the connection concept with stationary devices, we utilize Dyson hot+cool fans to provide heat and wind feedback. Both sensations from this fan are non-contact, thus the effect will decrease by the distance. To enhance the immersive experience through the tactile illusion effect, we design four interaction techniques as Figure 1. The particle effect design is depicted in Figure 3, where the green/red line represents the direction of the left/right controller. The gray circle denotes the effective area, and the blue line denotes the movement direction of the particle. The red circles are the final effective area surrounded by particles. Additionally, to augment wind and thermal feedback

from the stationary device, we dynamically adjust the feedback strength from the controller based on the distance between the device and the controller. Consequently, users could perceive the combination of the wind and thermal (cold/heat) feedback in VR.

3 APPLICATION: FANTASY WORLD

Particle effects and Sylph can be seen in many movies, animations, and games. To demonstrate the feasibility of our concept, we create an immersive air-bender experience with haptic feedback. In this demonstration, our system simulates a hot summer and a cold winter. At the beginning of the summer scene, users can perceive only the wind feedback. When the sun emerges, both wind and warmth feedback are provided. In the winter scene, users perceive the wind and cold feedback simultaneously. By clicking the trigger on the controllers, users can interact with the wind particles in the environment, bending the wind between their hands. As they create these visual effect particles, users simultaneously generate haptic wind and thermal feedback for a more immersive interaction. This fantasy world could be a multi-sensory environment to enhance the gaming and story narration experience.

4 DISCUSSION AND FUTURE WORK

We present a hybrid-haptic system with particle effect, creating a continuous illusion of wind from both stationary and portable devices to augment the sensation of wind in VR. The particle effect connects sensations from our device and the fan, resulting in an enhanced immersive experience through complementary wind feedback. In the future, the concept presented in this work also could potentially extend to a new kind of artistic experience—drawing with color particles that evoke both wind and thermal feedback. We will collect user feedback to inform researchers, game makers, and designers about the utility of this technique.

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